

Data Science

CS301

Multiple Linear Regression Understanding the Summary

Week 10

Fall 2025

Oliver BONHAM-CARTER



Up To Now in Regression

- We have discussed how one entity influences another.
- What about having two entities (independent) which may have some kind of influence on a dependent variable.
- Especially if a dependent variable has a high correlation with more multiple independent variable.



How is Grade Point Average Related to Time Studying?

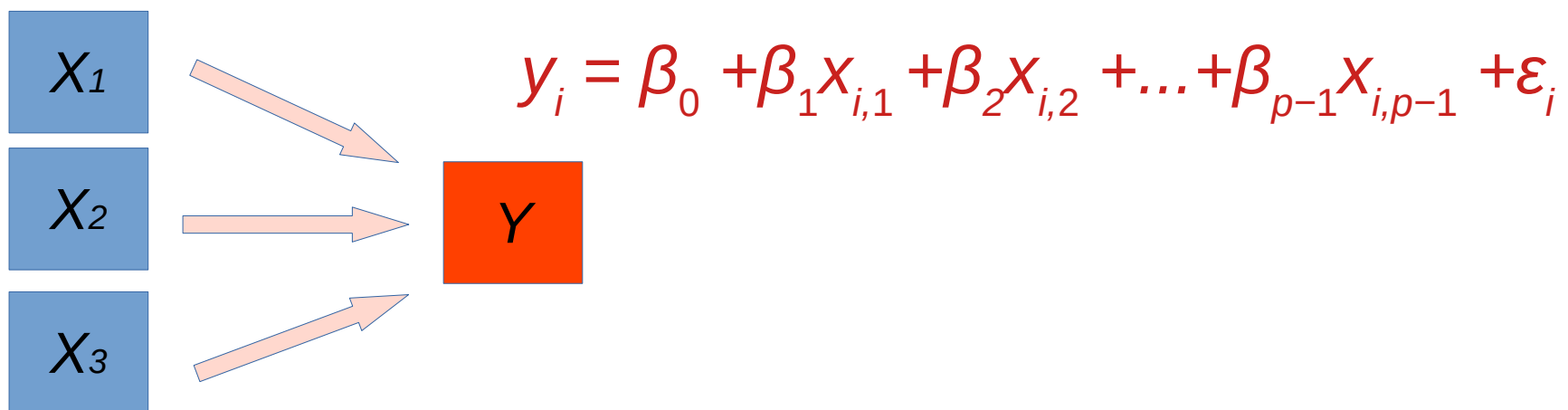
- “GPA could be dependent on studying”
 - *Student performance may be based on more than just one entity.*
- “GPA could be dependent on studying AND getting enough rest”
 - *Or maybe even more variables are involved?*
- GPA could be dependent on studying AND rest AND eating healthy food AND ... ??
 - *How do I begin to study this?*

So, Multiple Linear Regression Is What ... ?

Simple regression considers a single explanatory (independent variable) and a response (dependent) variable



Multiple regression simultaneously considers the influence of multiple explanatory (independent variables) on a response (dependent) variable





Types of Questions to Address

- Do **age** and **IQ** scores effectively predict *GPA*?
- Do **weight**, **height**, and **age** explain the variance in *cholesterol levels*?
- Are **elevated video game sales** explained by their exciting **graphics** and **inexpensive costs**?
- Is road **safety** a combination of **active** and **defensive** driving?
- Are there more independent variables to be studied with these dependents?



Equation of Multiple Independent Variables

- The model is now a multi-independent variable equation.

y_i  Dependent Variable

$$= \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,3} + \epsilon_i$$

 Independent Variables



Hypotheses

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$$

(Null Hyp)

$$H_0 : \beta_1 = \beta_2 \dots \beta_{k-1} = \beta_k = 0$$

(Alt Hyp)

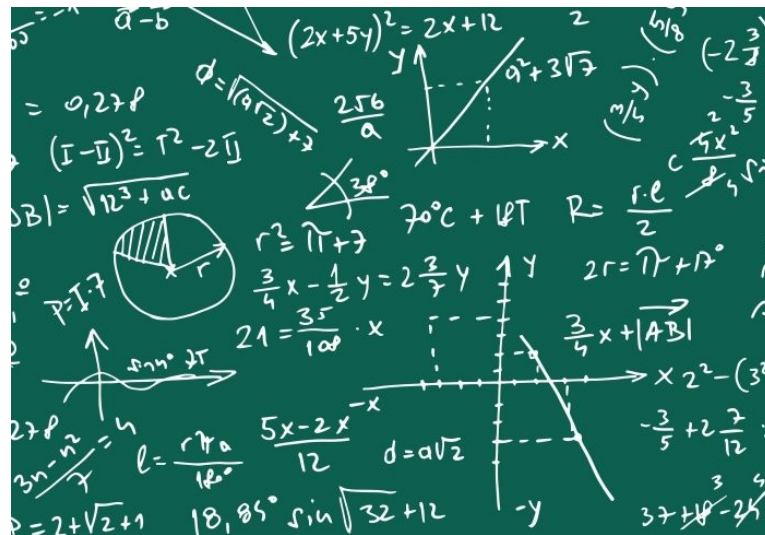
$$H_A : \text{Not all } \beta\text{'s} = 0$$

- The null hypothesis of a multiple regression is that there is no relationship between the indep variables and the dep variables (e.g., all regression coefficients are zero)
- The alternative hypothesis is that at least one coefficient is not zero.

Analysis Question

- Two variables: Do *Age* and *Height* (both) influence the capacity of lungs (*LungCap*)?
- Asking actually, can we make a model that takes the following form?

$$\text{LungCap} = \text{Age} * b_1 + \text{Height} * b_2 + b_3$$



See [sandbox/lungCap/lungCapDemo.R](#) for code



Lung Capacity Data

```
library(tidyverse)
# install.packages("psych")
library(psych)

#open lung capacity data
lc <- file.choose()
dataLungCap <- read.csv(lc, sep = ",", header = T)
View(dataLungCap)
```

Create the Multiple-Variable Regression Model

```
# model creation
```

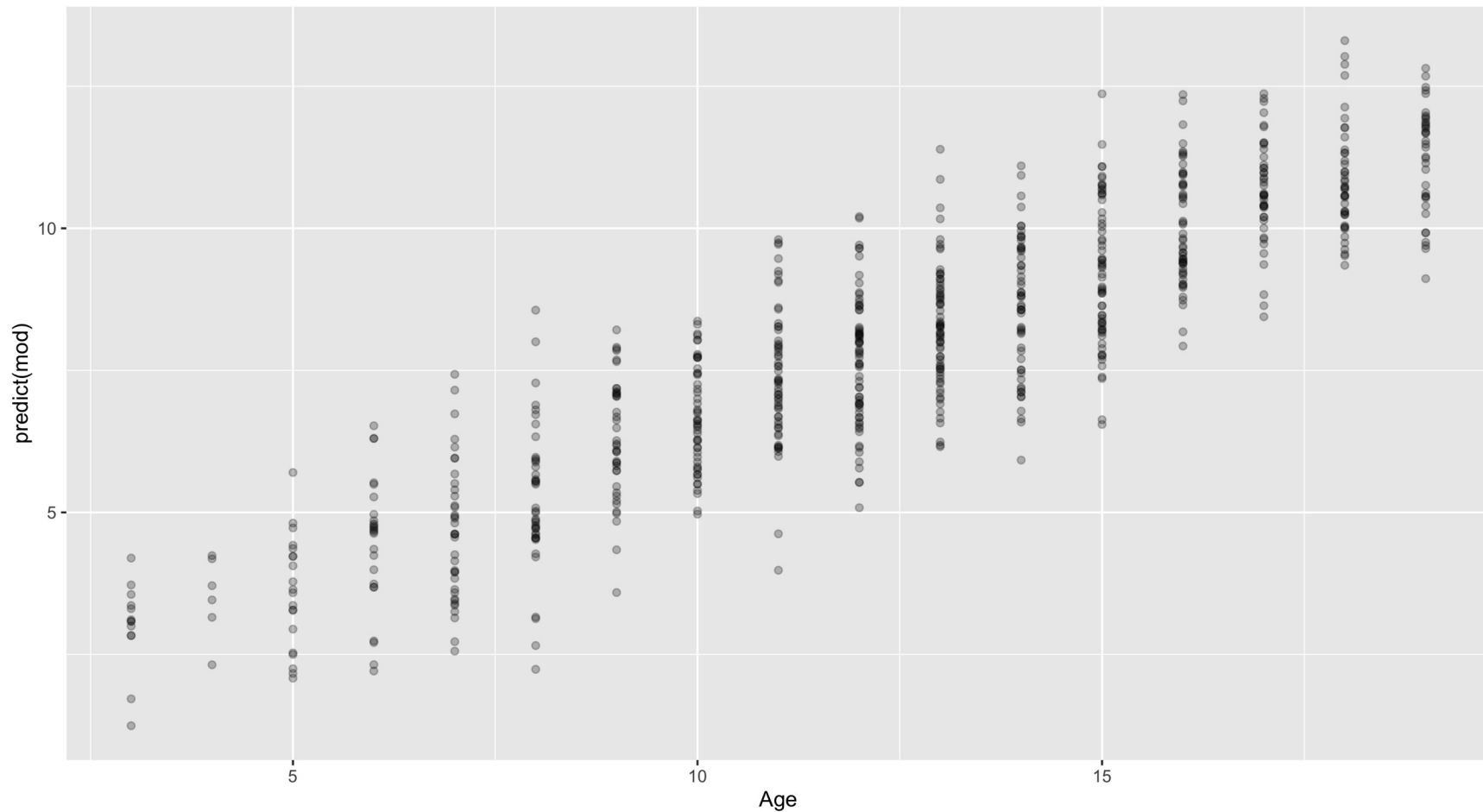
```
mod <- lm(data = dataLungCap, LungCap ~ Age +  
Height)
```

```
# get a report of the model
```

```
summary(mod)
```



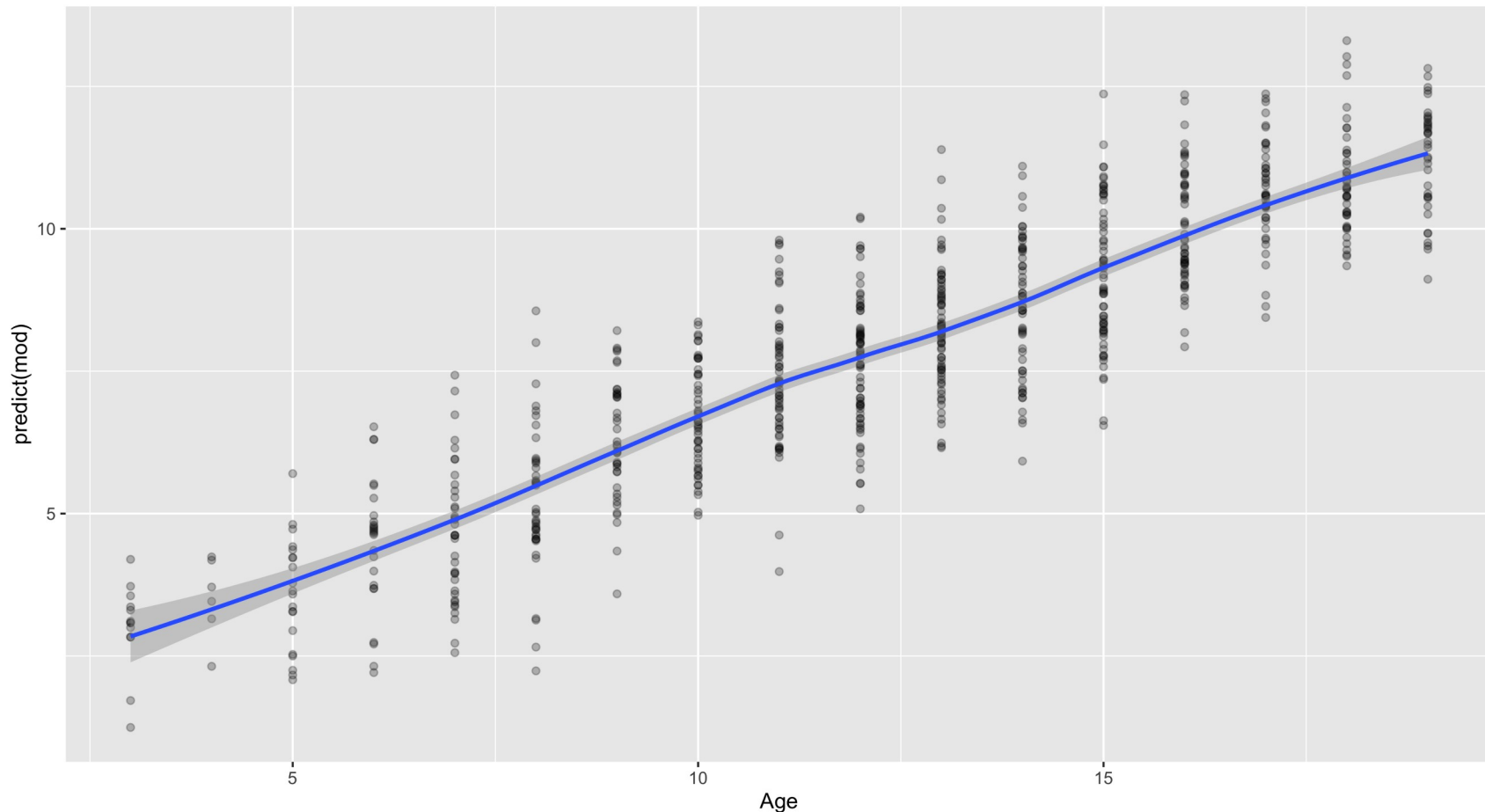
Scatter Plot



```
dataLungCap %>% ggplot(aes(x = Age, y =  
predict(mod))) + geom_point(alpha = 1(1/4))
```

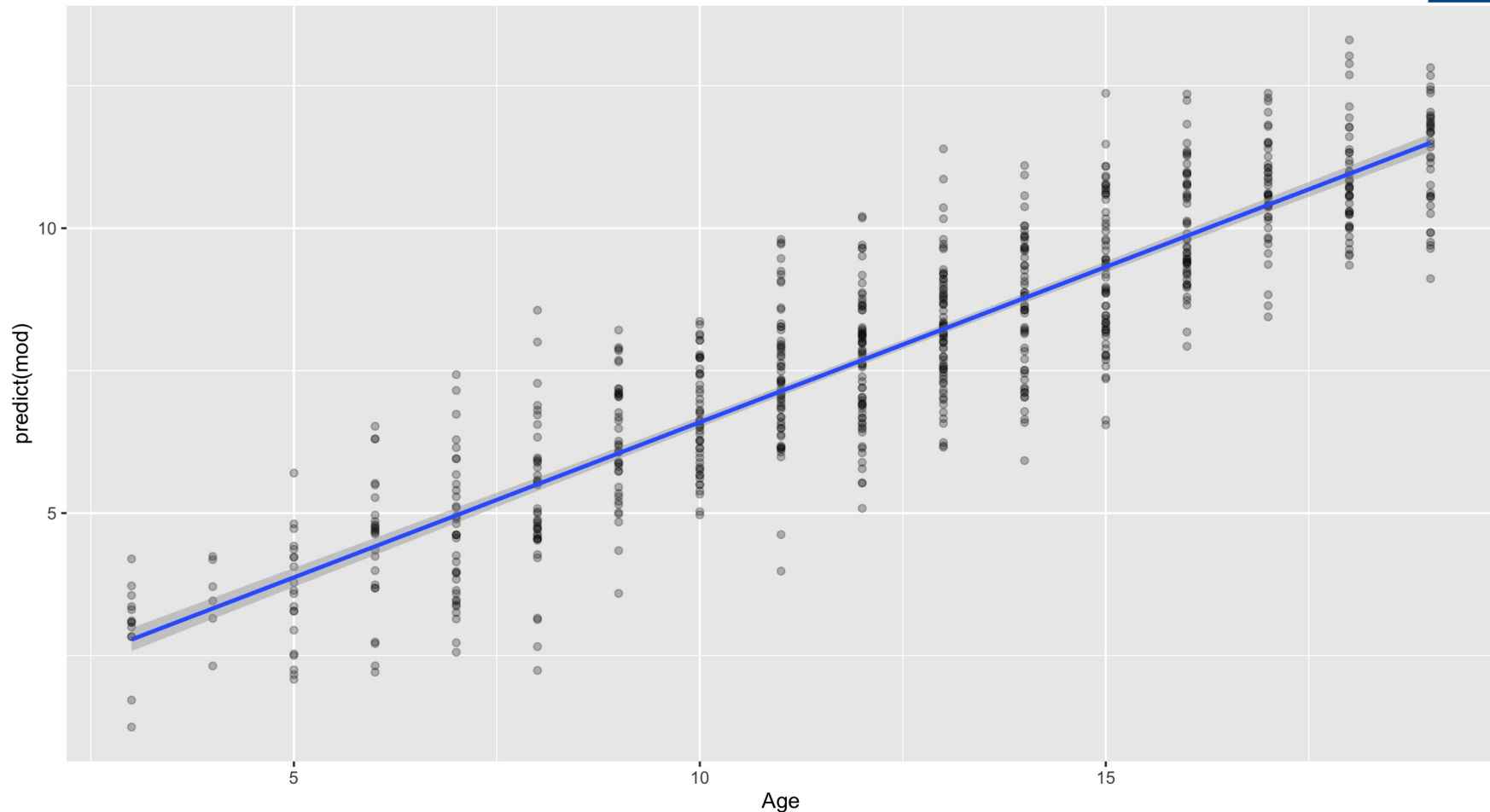


Plot With Smooth Line



```
dataLungCap %>% ggplot(aes(x = Age, y =  
predict(mod))) + geom_point(alpha = 1/4) +  
geom_smooth()
```

Plot With Model Line



```
dataLungCap %>% ggplot(aes(x = Age, y =
predict(mod))) + geom_point(alpha = 1/4) +
geom_smooth( method = lm)
```



Summary

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16



Intercept Value: “When the age and height are zero”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16 ***
Age	0.126368	0.017851	7.079	3.45e-12 ***
Height	0.278432	0.009926	28.051	< 2e-16 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The estimated mean lung capacity of someone having an age and height of zero. Is this *meaningful*?



Slope of Age:

“How is my *Age* variable related to *Height*?”

Call:

```
lm(formula = LungCap ~ Age
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7100	3.4080

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The effect of *Age* on *Lung Capacity* adjusting or controlling for *Height*. We may associate an increase of 1 year in *Age* with an increase of 0.126 in *Lung Capacity* adjusting or controlling for *Height*



Test Statistic

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The test statistic that we use to perform the hypothesis test that the slope for Age = 0.



Slope of Height:

“How is my *Height* variable related to *Age*?”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476855	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The estimated
effect of *Height* on
Lung Capacity,
adjusted for *Age*.



Test Statistic:

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The test statistic that we use to perform the hypothesis test that the slope for *Height* = 0.



R-squared Value:

“How do the independents explain the dependent?”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	(> t)
(Intercept)	-11.747065	0.476899	-24.451	< 2e-16 ***
Age	0.126368	0.017851	7.079	3.45e-12 ***
Height	0.278432	0.009926	28.051	< 2e-16 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

Approximately 84% of the variation in *Lung Capacity* can be explained by our model (*Age* and *Height*)



Adjusted R-squared Value:

“How do the independents explain the dependent?”

Call:

lm(formula = LungCap ~ Age + Height)

**Approximately 84% of the
variation in *Lung Capacity*
can be explained by
our model (*Age* and *Height*)**

Adjusted R-squared is a modified version of R-squared value.

Value has been adjusted for the number of predictors in the model. The adjusted R-squared increases when the new term improves the model more than would be expected by chance.

It decreases when a predictor improves the model by less than expected.

**

**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16



F-Statistic of Test:

“What value do I look up in a table to check on significance?”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	-11.747065	0.476899	-24.63
Age	0.126368	0.017851	7.08
Height	0.278432	0.009926	28.01

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

Null Hyp. Test:

The test of the null hypothesis that all model coefficients are zero.

Degrees of Freedom:

There are 725 rows in the data and three groups.

$$722 = 725 - 3$$

Used also for finding values in F-table for hypothesis testing

Our Test of The Null Hypothesis

- **Ho:** $\beta_1 = \beta_2 = \dots = \beta_k$

Nothing is happening between the k-number of variables

- In our case,

- **Ho:** $\beta_{\text{age}} = \beta_{\text{height}} = 0$ (slopes are zero)

- **Ha:** $\beta_{\text{age}} \neq \beta_{\text{height}} \neq 0$ (slopes not all zeros)

$$y_i =$$

$$\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,3} + \epsilon_i$$



The p-Value: “Is this model statistically meaningful?”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1

Coefficients:

	Estimate	Std. Error	t	Pr(> t)
(Intercept)	-11.747065	0.476899	-24.4	***
Age	0.126368	0.017851	7.079	***
Height	0.278432	0.009926	28.051	***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

The p-value is very close to zero and so we reject the H_0 (i.e., all the model coefficients are zero (slope = 0)).

Conclusion: There is something non-random happening in this model.



Residual Errors:

“What is the estimation of the difference between observed and predicted values?”

Call:

```
lm(formula = LungCap ~ Age + Height)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1679

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16	***
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

This error gives an idea about how far the observed *Lung Capacity* (dependent) values are from the predicted or fitted *Lung Capacity* (the “y-hats”)

Residual standard error: 1.056 on 722 degrees of freedom

Multiple R-squared: 0.843, Adjusted R-squared: 0.8425

F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16



Assumptions: Linear Relationships

- A multiple regression model relating a y -dependent variable to p multiple independent variables

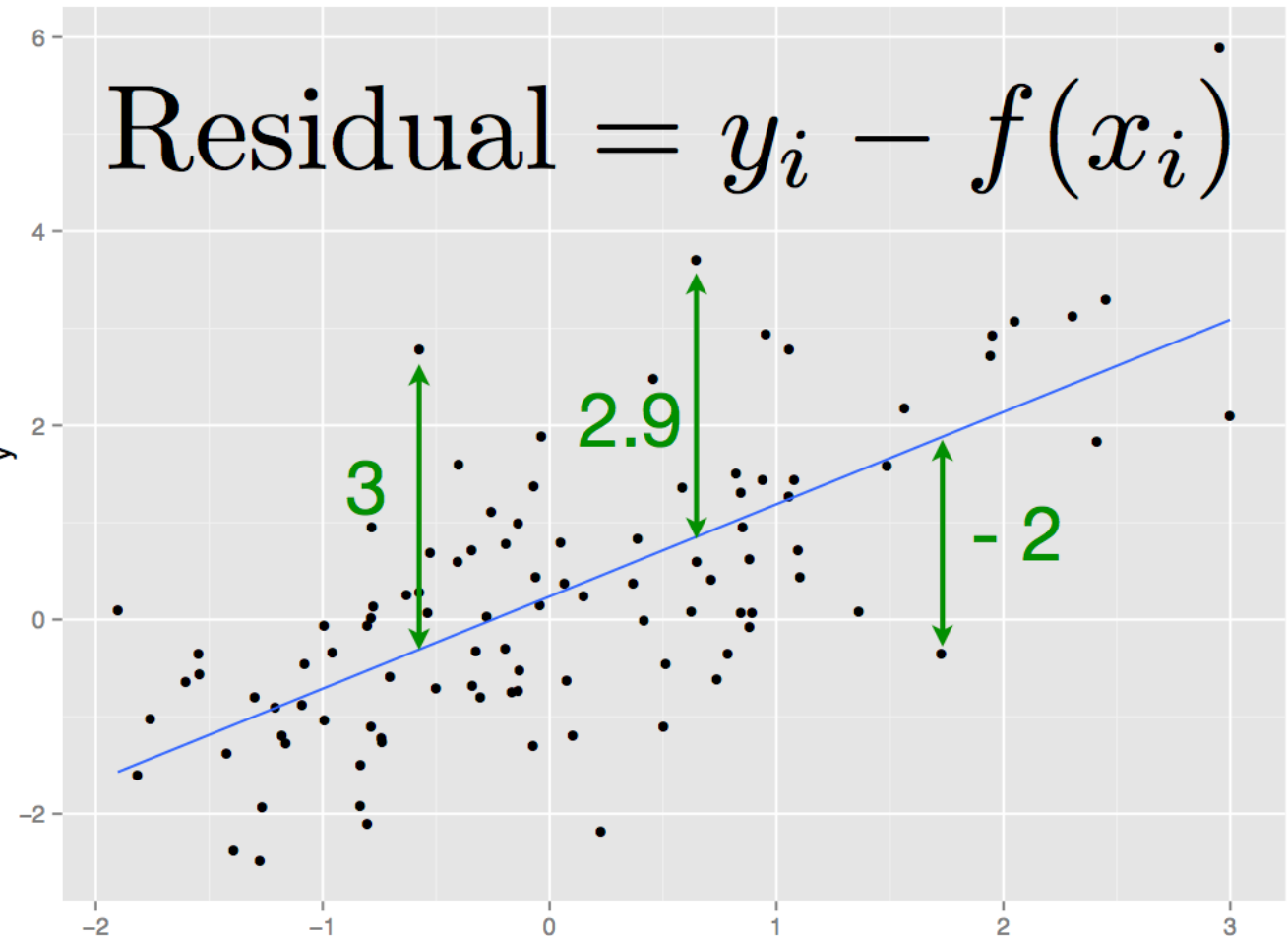
$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$$

- There must be a linear relationship between the independent variable and the independent variables.
 - Use a scatterplots to explore a linear or curvilinear relationship.

File sandbox/assumptions_multicollinearity.R

Assumptions: Residuals (From Last Time)

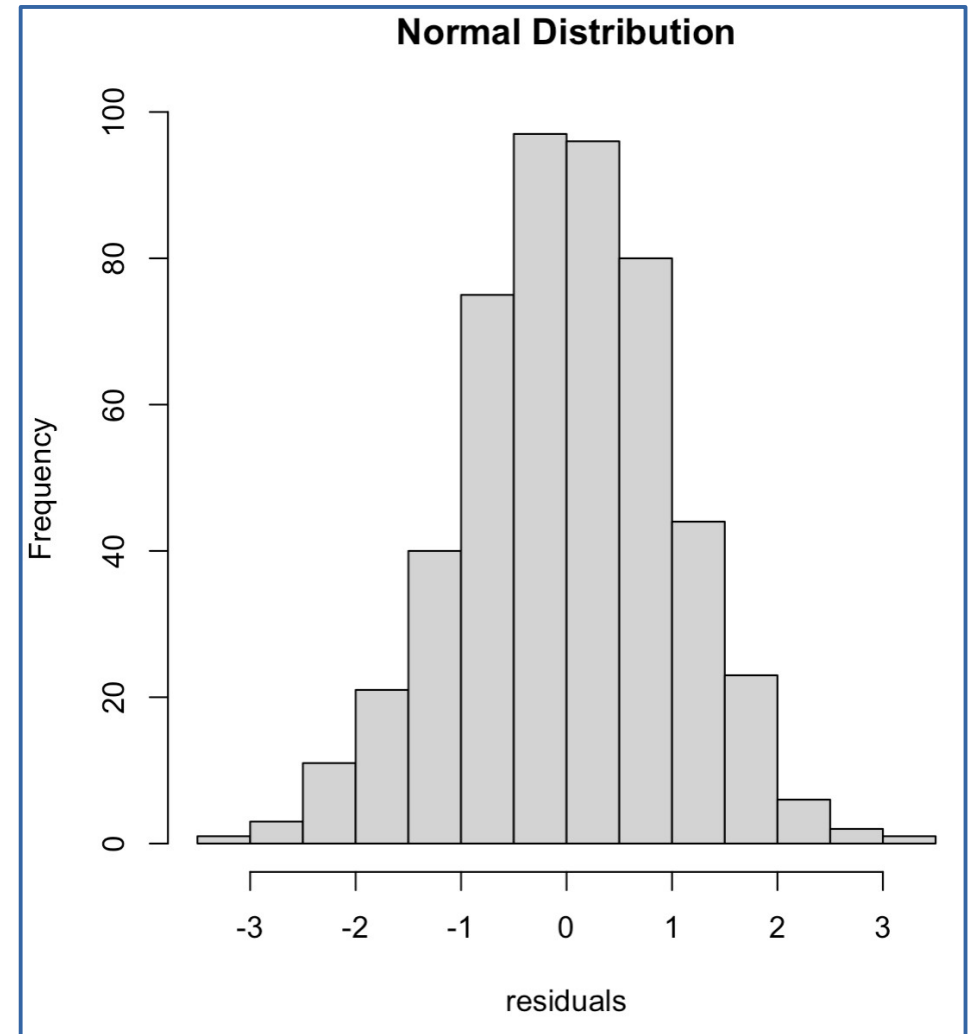
- A *residual* of an observed value is the difference between the observed value and the estimated value of the quantity of interest
- Are your residuals spread all over the line? No gaps?





Assumptions: Residual Normal Distribution

- Multivariate Normality—Multiple regression assumes that the residuals are *normally distributed*



If Variables Could Sing ...



Everyone's voice should have own *sound*?



Assumptions: No Multicollinearity

- Multicollinearity occurs when independent variables in a regression model are *correlated* (e.g., *indep vars are to be independent from each other.*)
- Multiple regression assumes that the independent variables are not highly correlated with each other. This assumption is tested using Variance Inflation Factor (VIF) values.
- Blog Post about VIF testing:
 - <https://www.r-bloggers.com/2023/12/exploring-variance-inflation-factor-vif-in-r-a-practical-guide/>

Test for Correlation Between *All* Variables

```
library(psych)  
lc <- file.choose()  
dataLungCap <- read.csv(lc, sep = ",", header = T)  
pairs.panels(dataLungCap)  
pairs.panels(dataLungCap, lm = TRUE)
```

Allows for a quick study
of correlation across
the variables of your
study

pairs.panels {psych}

R Documentation

SPLM, histograms and
correlations for a data matrix

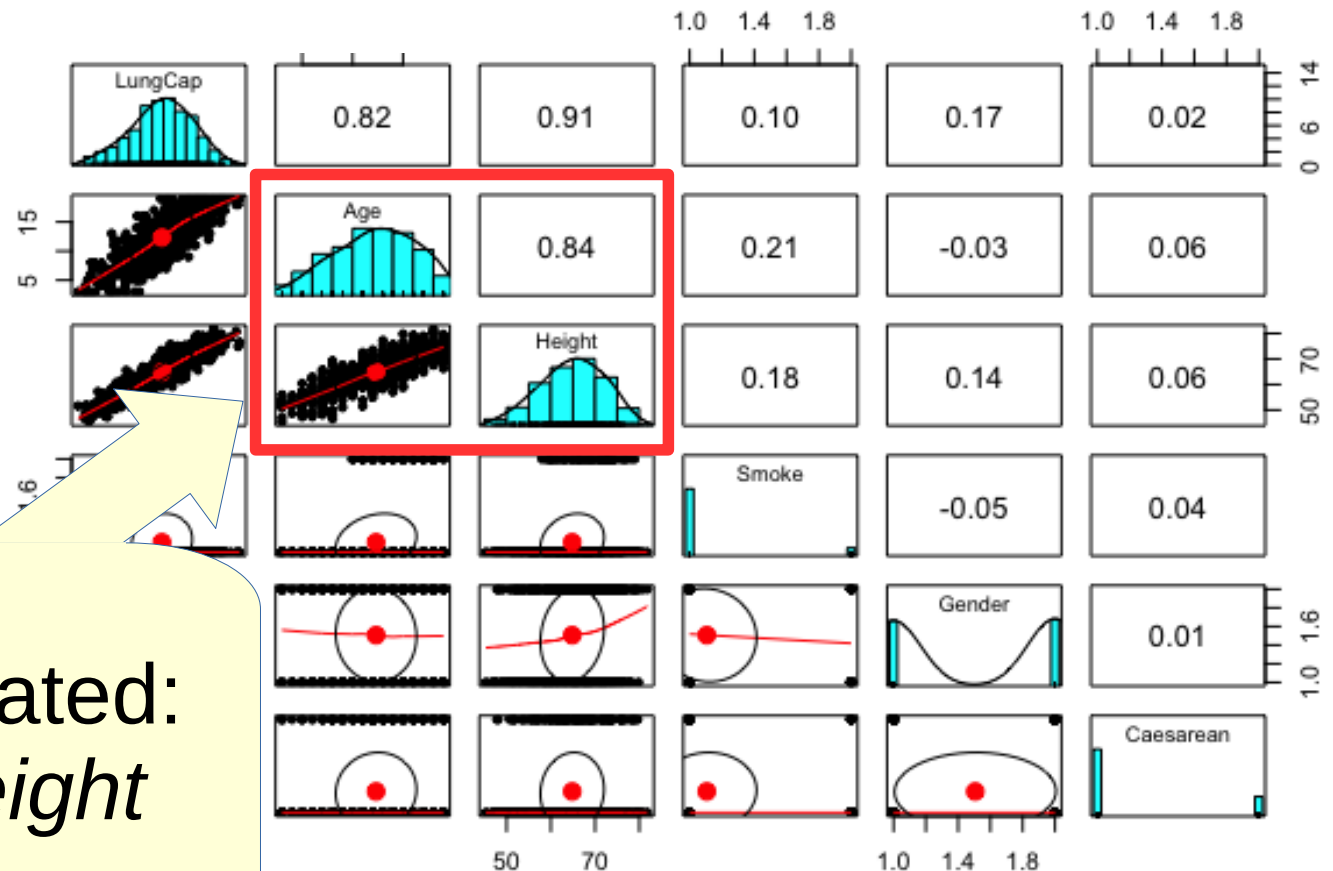
Description

Adapted from the help page for pairs, pairs.panels shows a scatter plot of matrices (SPLM), with bivariate scatter plots below the diagonal, histograms on the diagonal, and the Pearson correlation above the diagonal. Useful for descriptive statistics of small data sets. If lm=TRUE, linear regression fits are shown for both y by x and x by y. Correlation ellipses are also shown. Points may be given different colors depending upon some grouping variable. Robust fitting is done using lowess or loess regression. Confidence intervals of either the lm or loess are drawn if requested.

Correlation Between *Age* and *Height*

- Pearson correlation between Age and Height = 0.84

```
> cor(Age, Height, method = "pearson")  
[1] 0.8357368
```

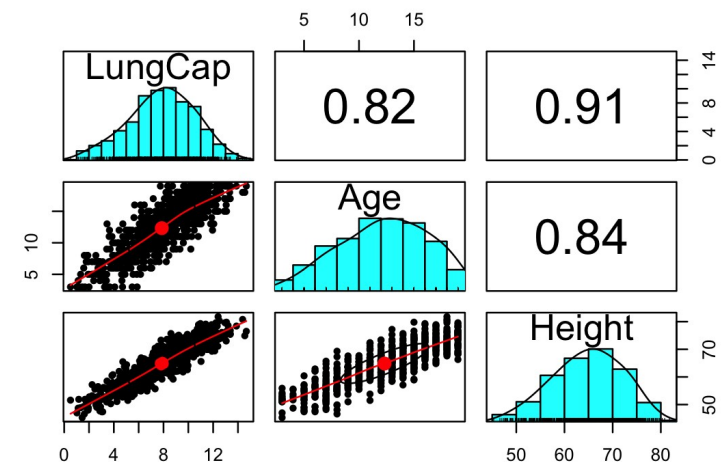
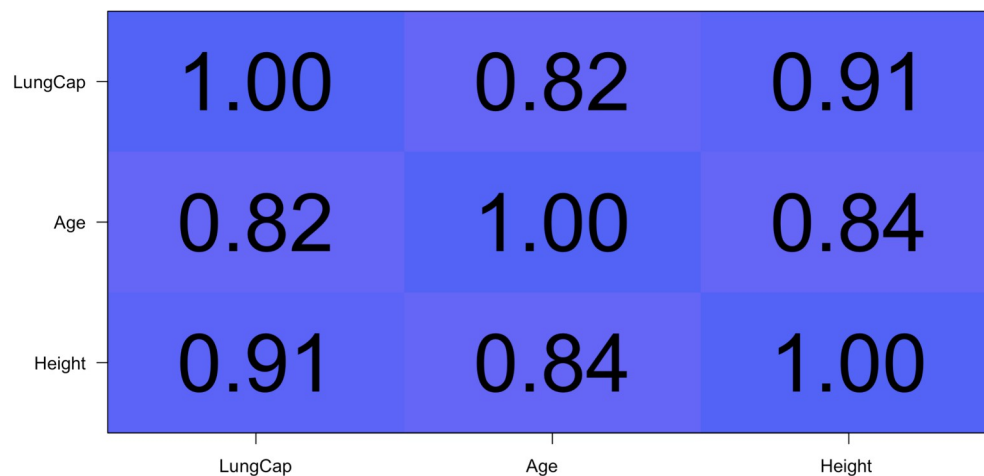


Highly correlated:
age and *height*

Test for Specific Correlations

```
#open lung capacity data
cor(dataLungCap$Age, dataLungCap$Height)
corPlot(dataLungCap[1:3])
pairs.panels(dataLungCap[1:3])
# output: 0.8357368
```

Correlation plot from data





Correlation and Confidence

(Remember that we are studying variable slope)

```
# Pearson correlation test  
cor(dataLungCap$Age, dataLungCap$Height)  
  
# output: 0.8357368  
  
# Examine the 95 percent confidence level  
confint(mod, conf.level = 0.95)
```

Default lower and upper confidence limits: 2.5% and 97.5%

The **estimated slope** for Age is 0.126 and we are 95 percent sure that the **true slope** of Age is between 0.09 and 0.16.

```
> confint(mod, conf.level = 0.95)
```

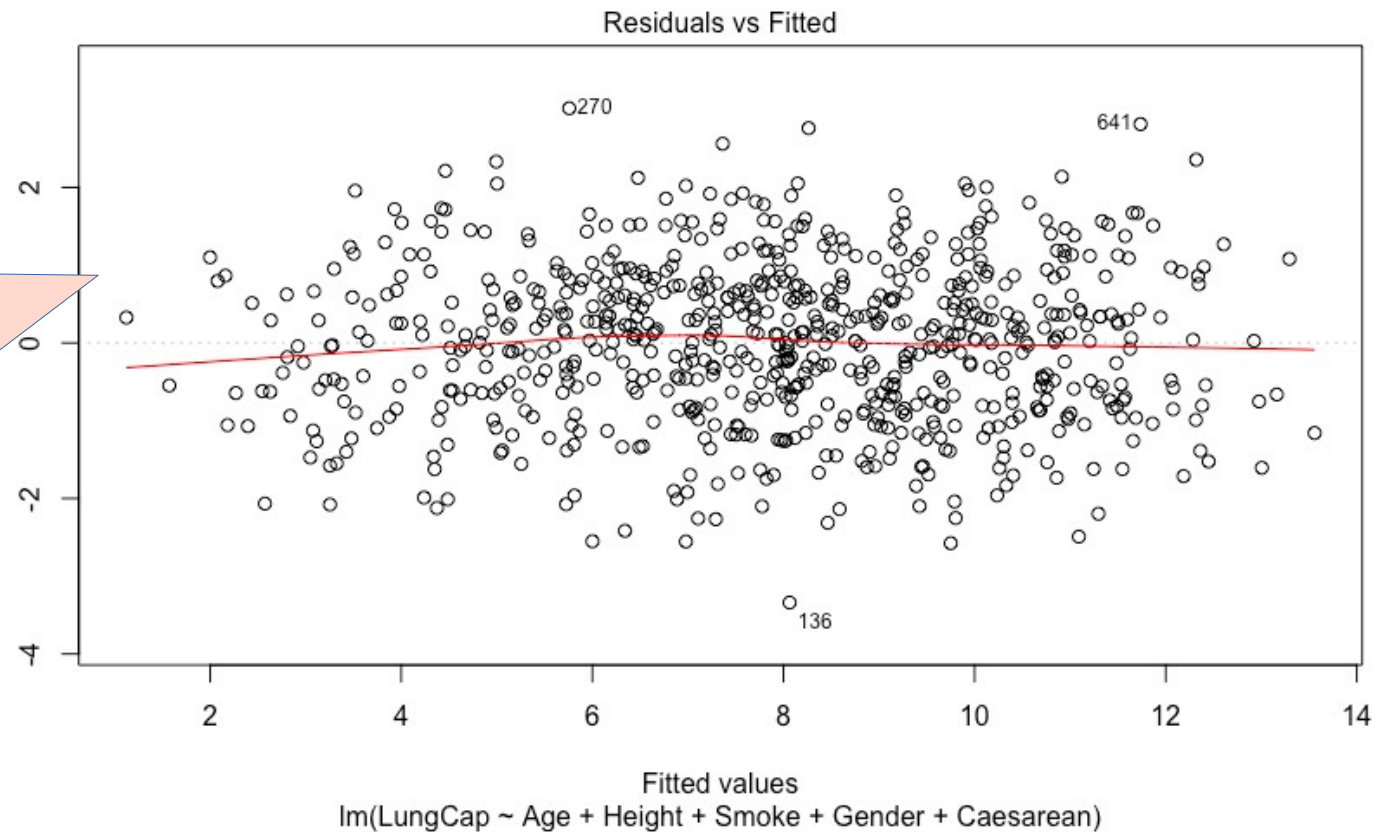
	2.5 %	97.5 %
(Intercept)	-12.68333877	-10.8107918
Age	0.09132215	0.1614142
Height	0.25894454	0.2979192



Create a Bigger Model!!

- `mod2 <- lm(data = dataLungCap, LungCap ~ Age + Height + Smoke + Gender + Caesarean)`
- `summary(mod2)`
- `plot(mod2)` # check the four plots!

Residuals Vs. Fitted:
The relationship
between *Age*,
Height and
Lung Capacity
is approx linear.





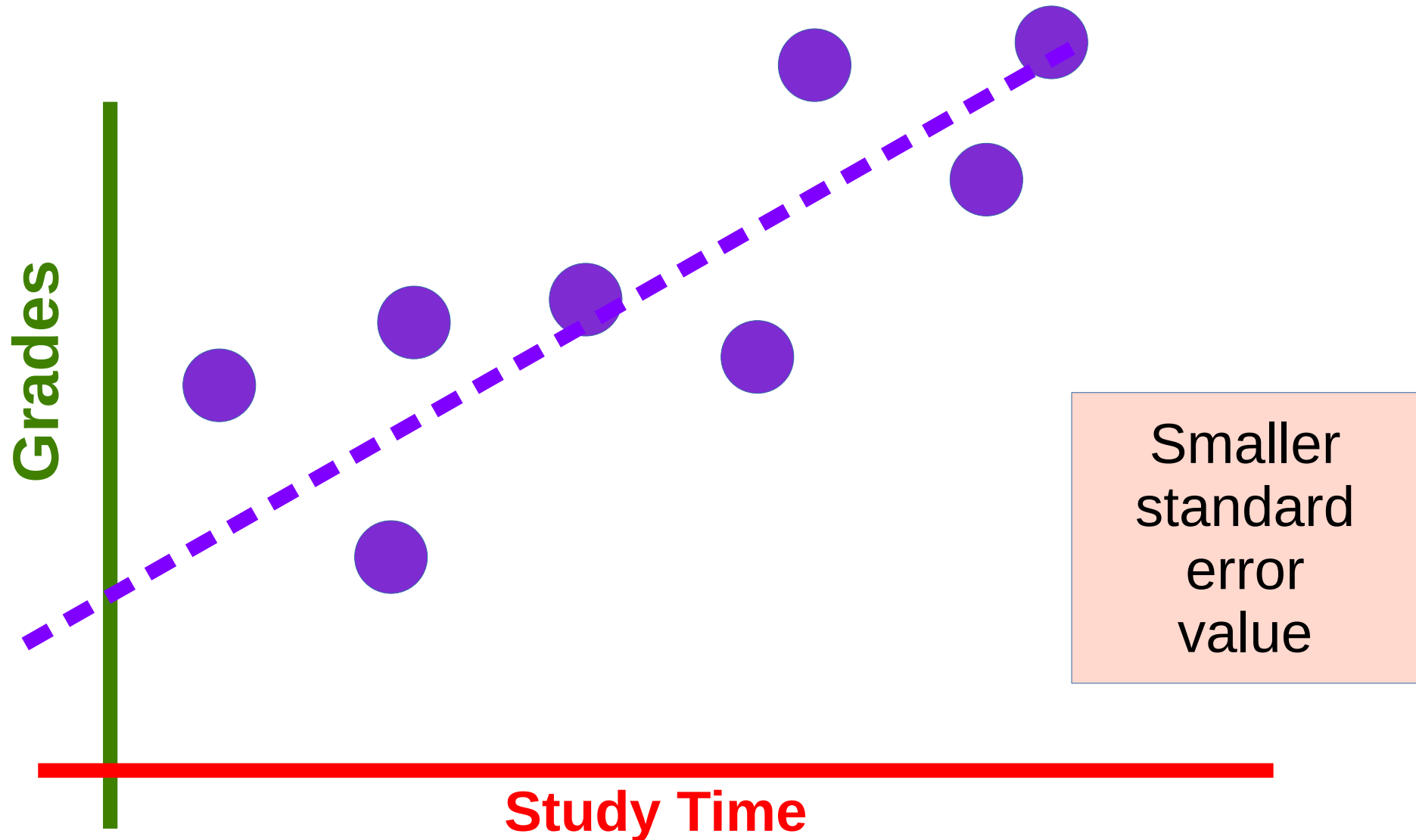
ALLEGHENY
COLLEGE

Lemme Say Something More About Residuals!!



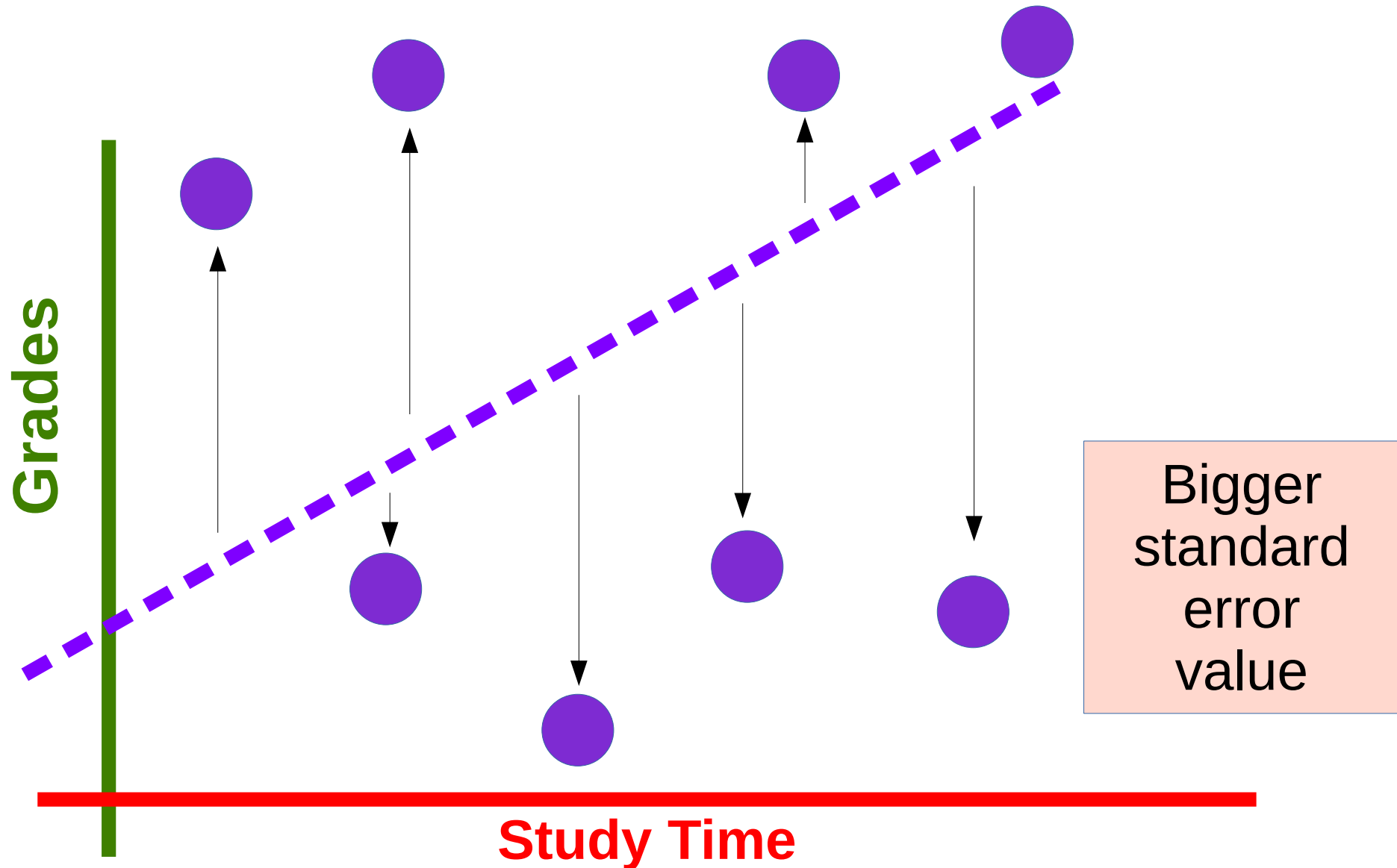


Assumptions: “Good” Residual Standard Error

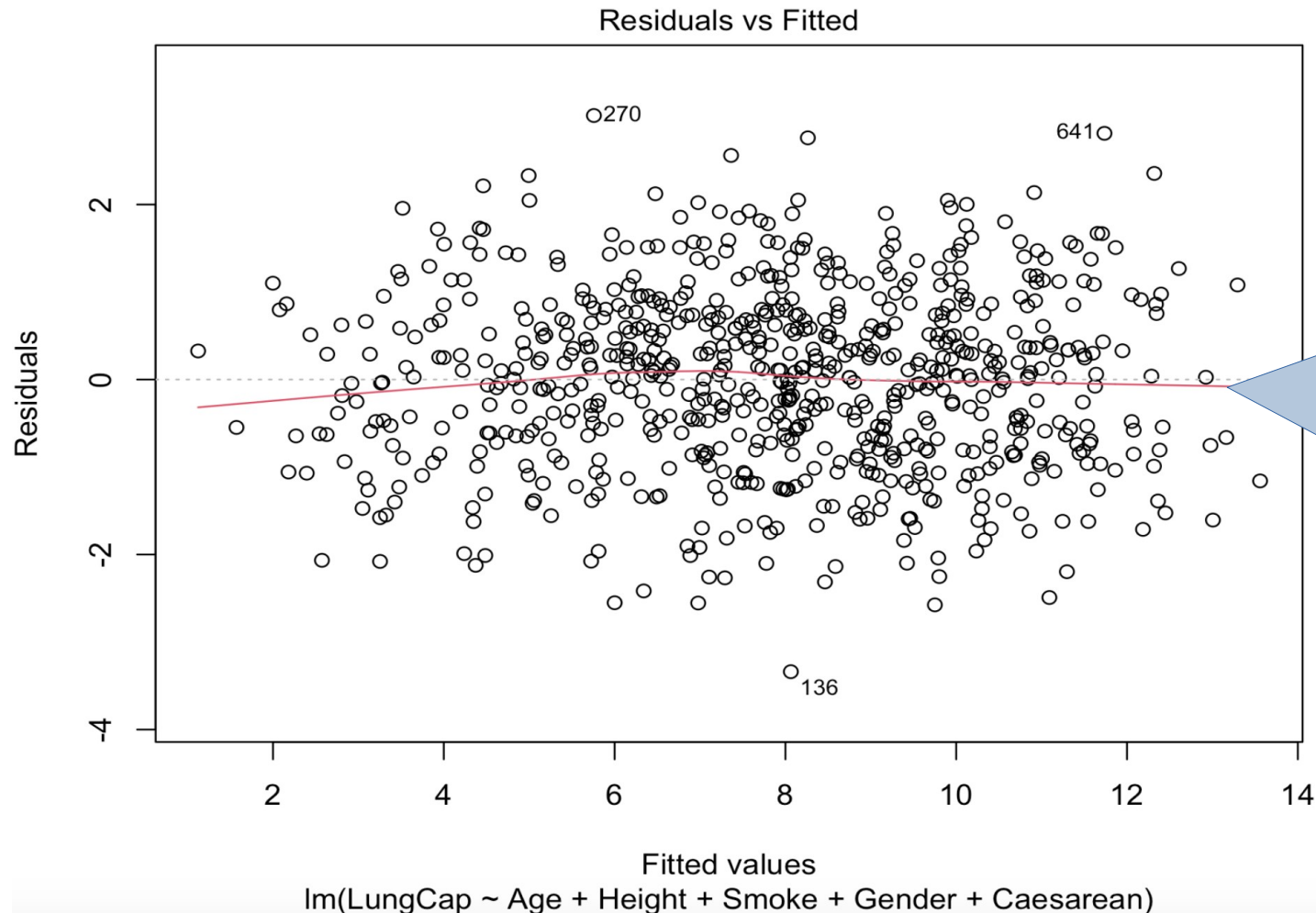




Assumptions: “Bad” Residual Standard Error



Residuals Vs. Fitted

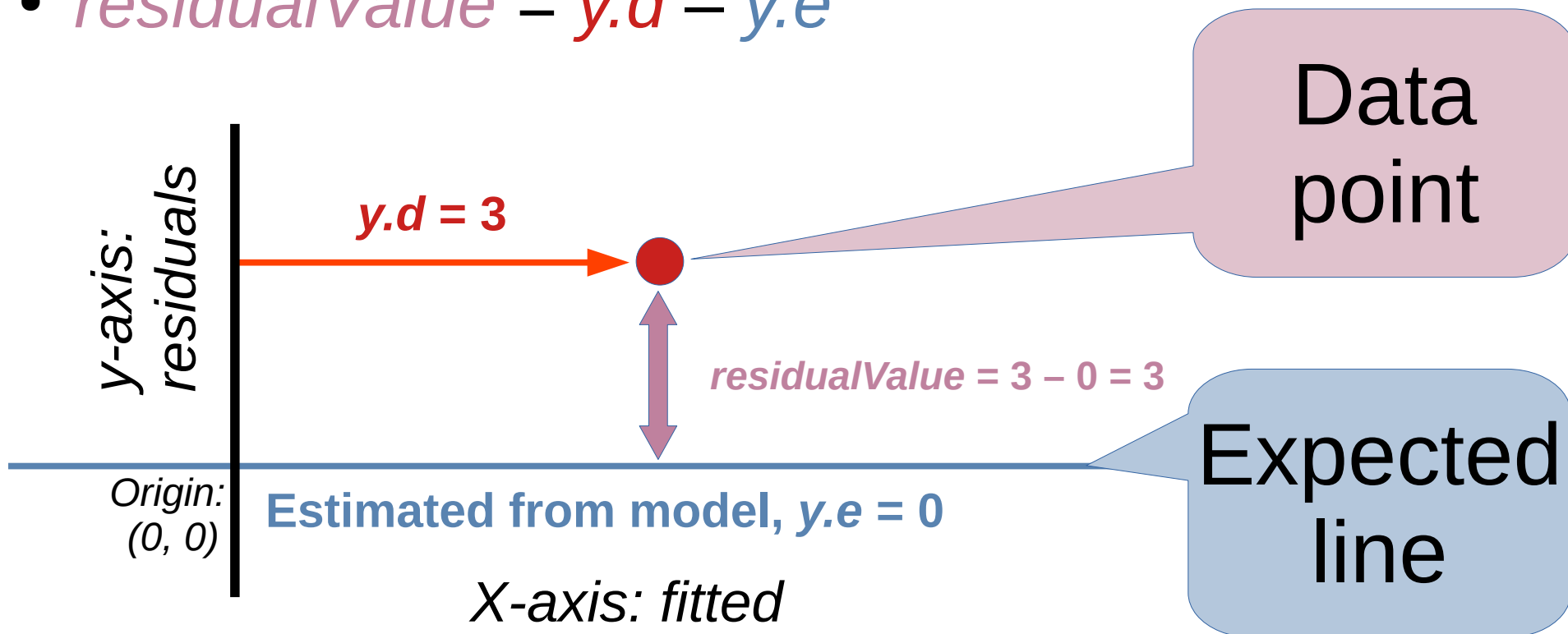


Expected
Line
(fitted)

- Points are vertical distances between $y.d$ (i.e., a datapoint) and $y.e$ (i.e., an estimated point)

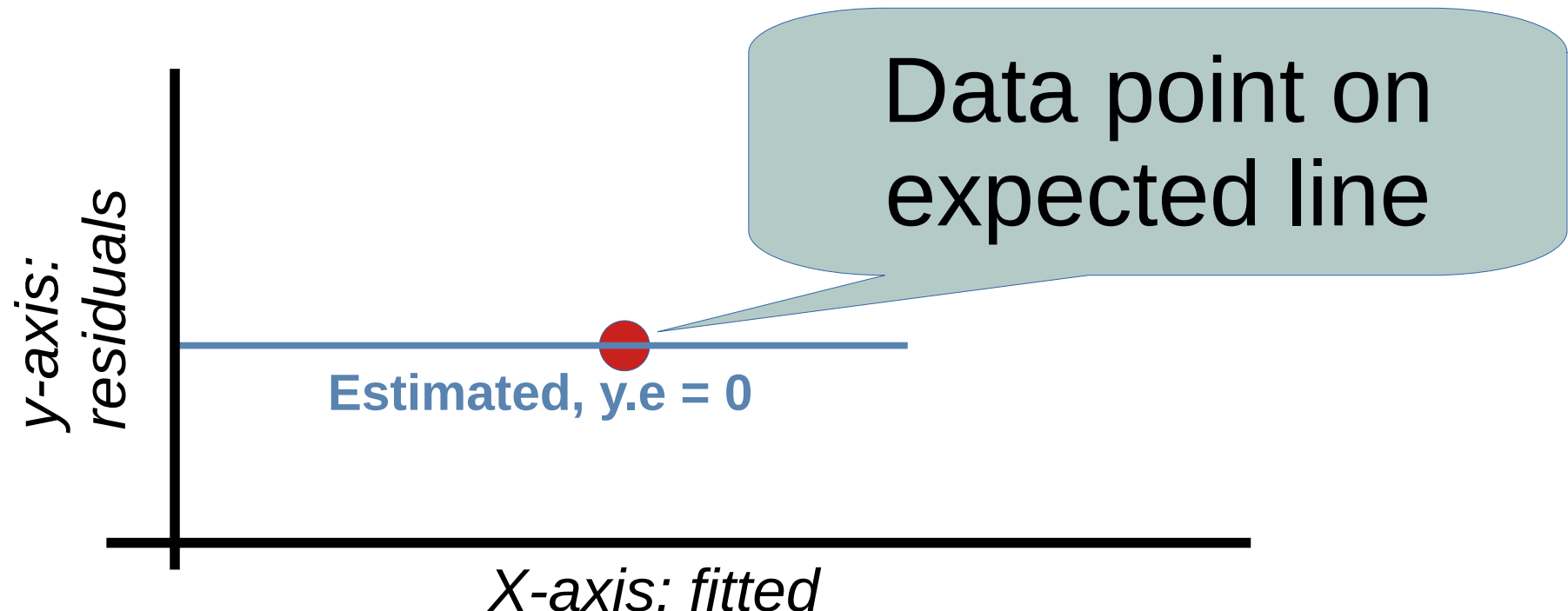
Residuals Vs. Fitted

- **Residual value:** a vertical distance between any one data point $y.d$ (i.e., datapoint) and the estimated value $y.e$ (i.e., estimated)
- $residualValue = y.d - y.e$

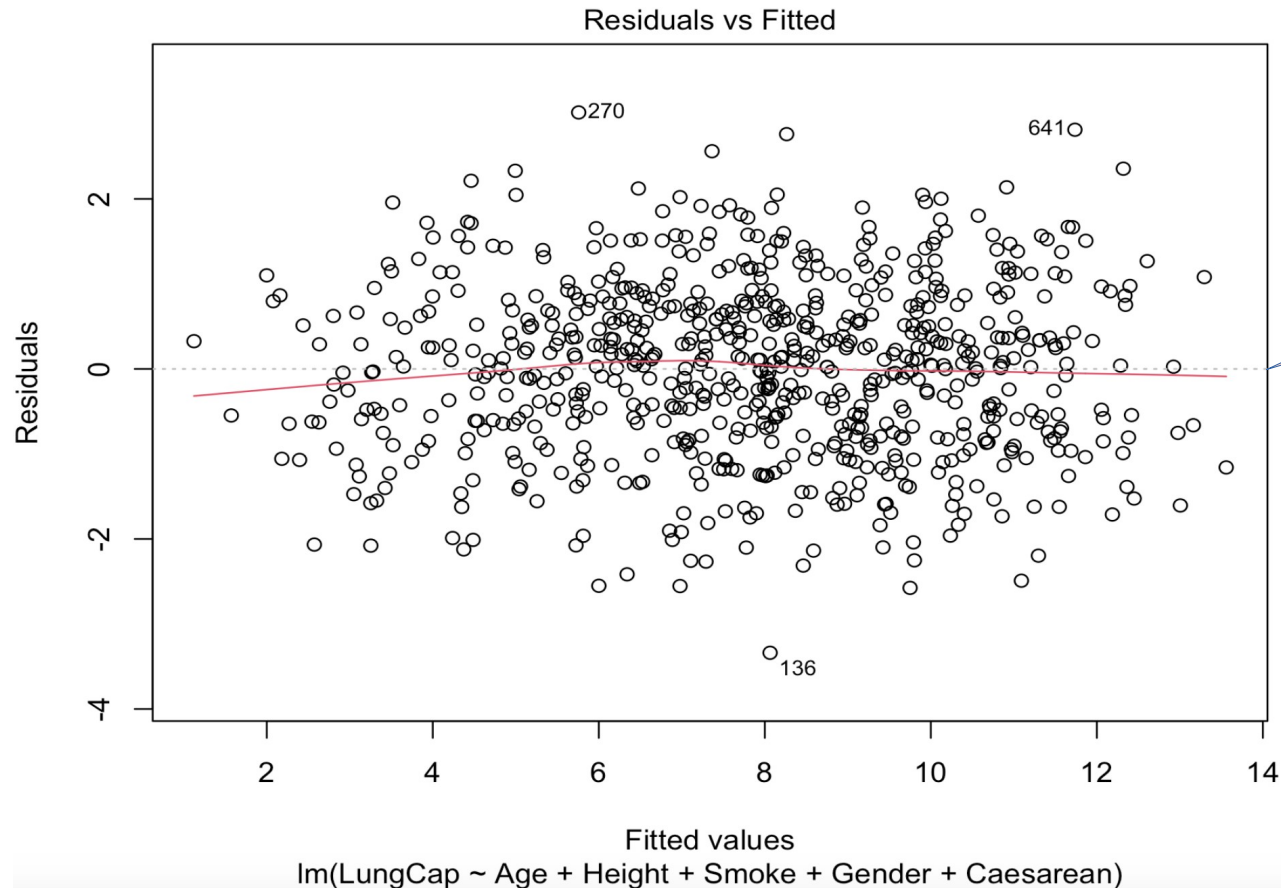


Residuals Vs. Fitted

- Any data point found (directly) on the estimated regression line has a residual of 0
- The residual = 0 line corresponds to the estimated regression line (e.g., the data point is on the model)
- Model is appropriate (accurate) for data point

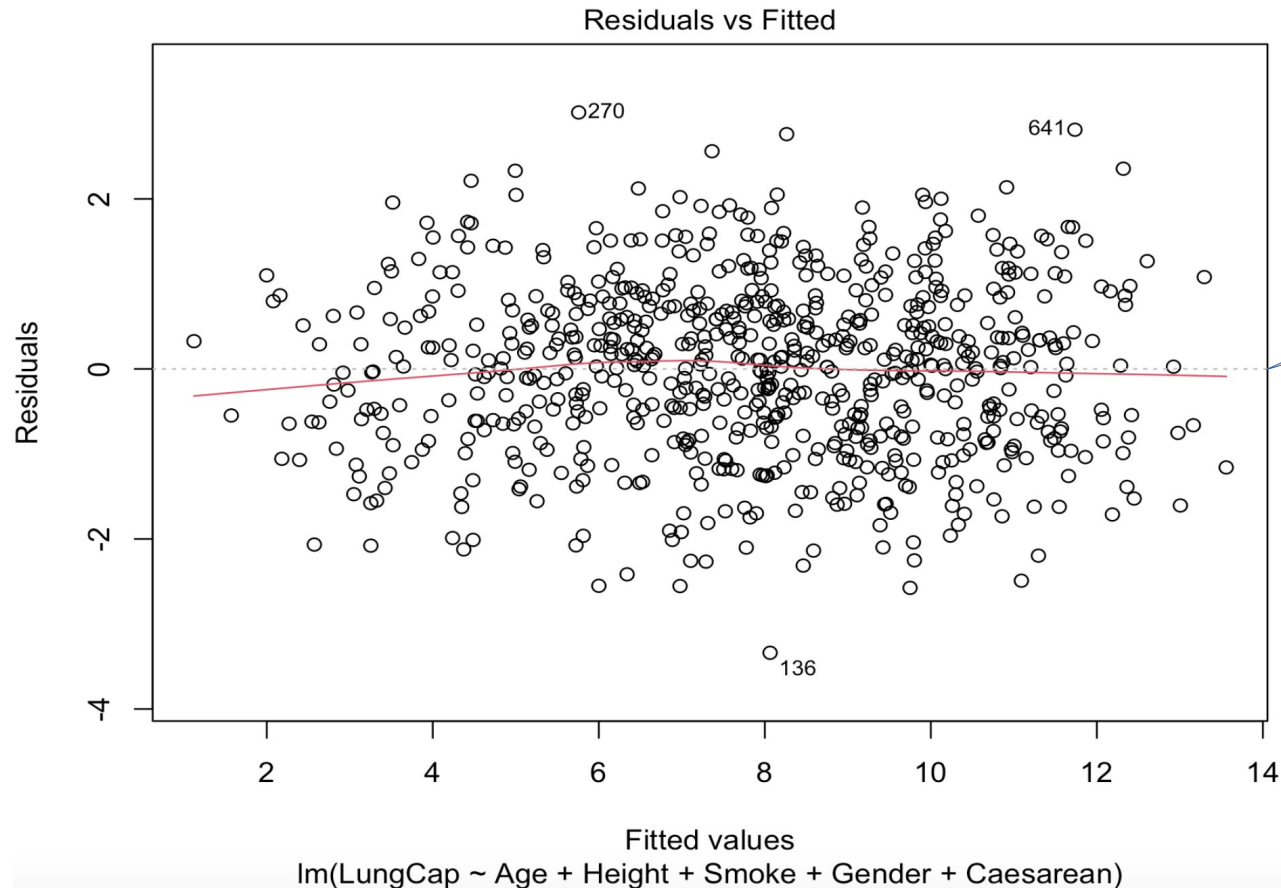


First Thing the Plot Suggests



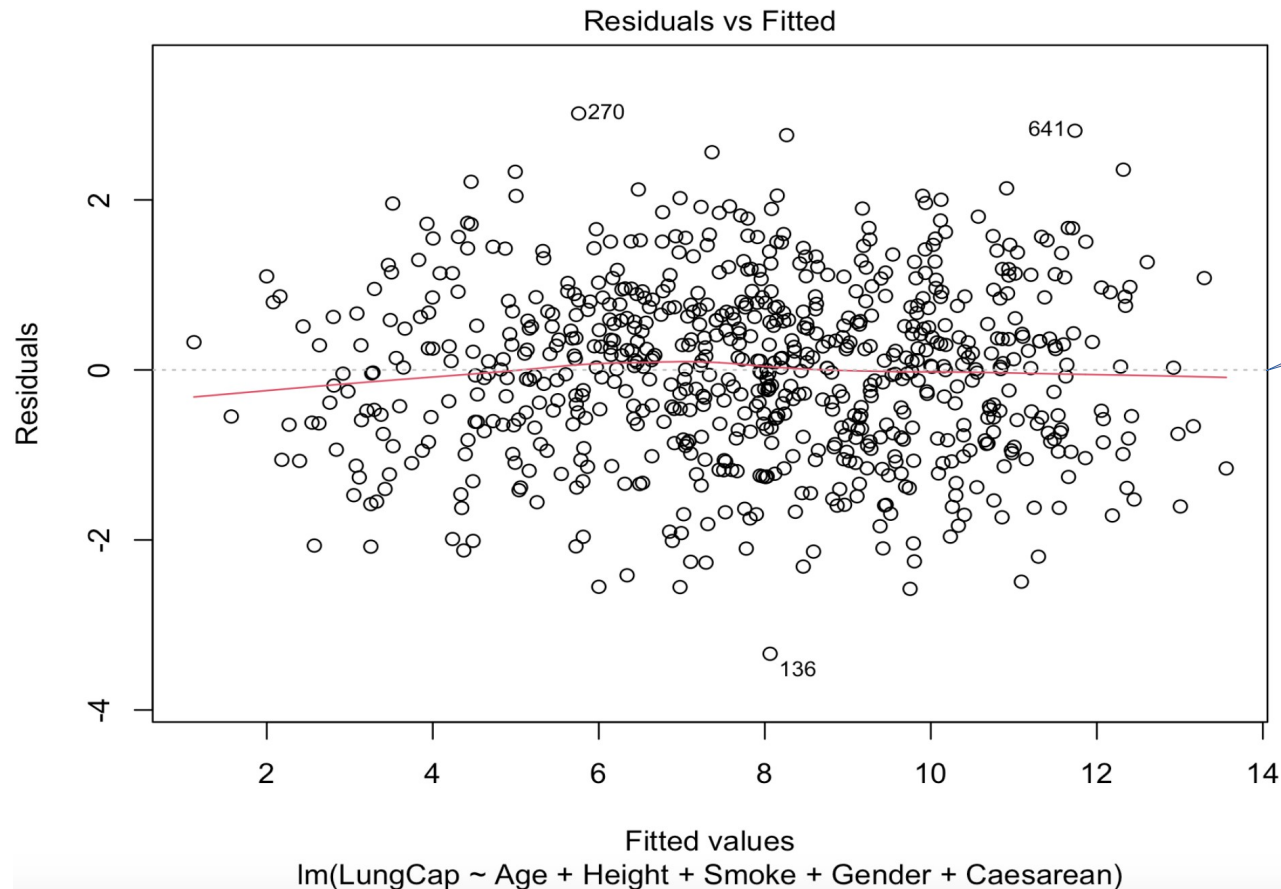
- The residuals are generally (randomly) situated around the 0 line.
- This suggests that the assumption that the relationship is linear is reasonable.

Second Thing the Plot Suggests

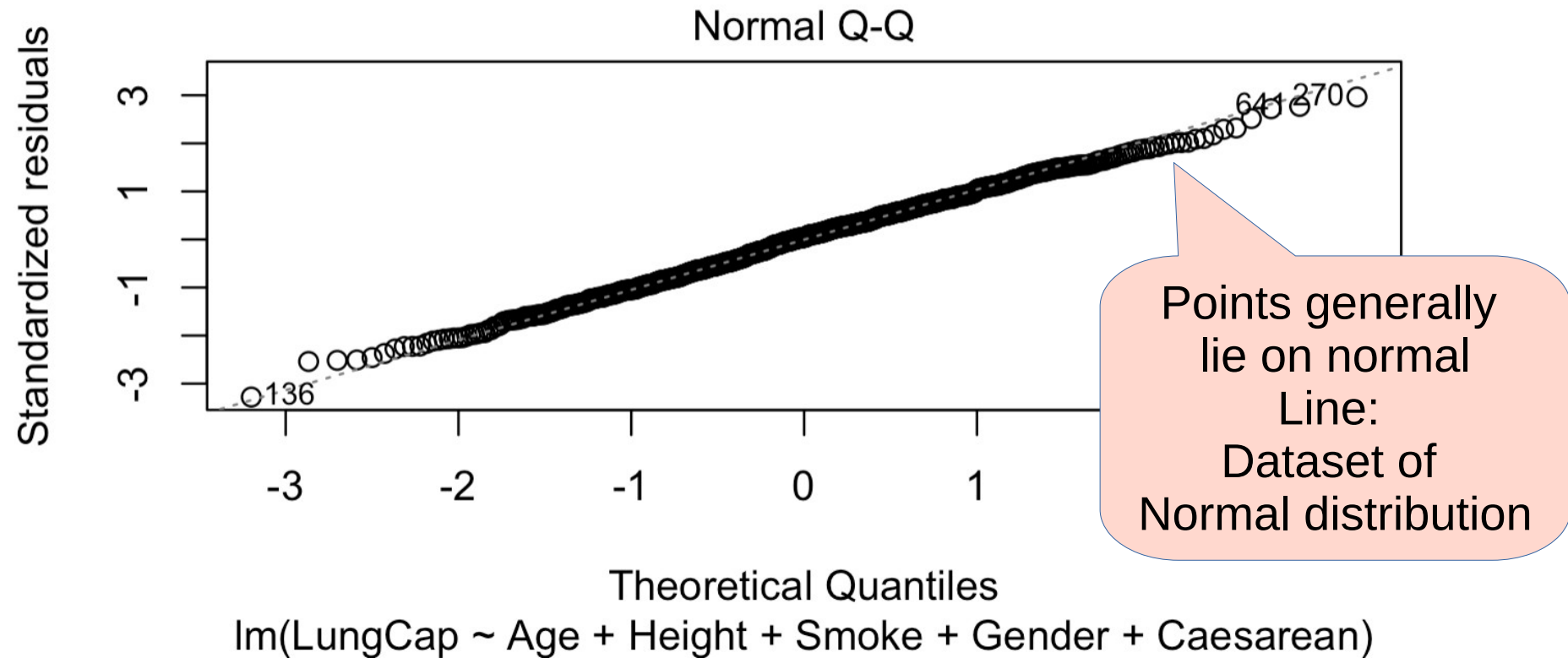


- The residuals roughly form a "horizontal band" around the 0 line.
- This suggests that the variances of the error terms are equal.

Third Thing the Plot Suggests



- Few residuals are "standing out" from the basic random pattern of residuals.
- This suggests that there are few outliers (since few large distances between expected line and data point)



- Residuals aligning diagonal line suggests dataset comes from normal distribution
- Outliers/Tails: An "S" shape can indicate more extreme values (i.e., *outliers*) than expected, while a straighter line in the center with little deviation at the ends suggests a better fit.

Go Create Your Own Model!!

- Find another dataset to make a model
- Fit a linear model using ALL x variables.

